

1. System Functionality

SYS-001

Description: The system shall provide the capability to input signal values for speed, temperature, battery level, and left, right, and warning blinker status through the server GUI.

SYS-002

Description: The system shall provide the capability to display signal values for speed, temperature, battery level, and left, right, and warning blinker status through the client GUI.

SYS-003

Description: The system shall provide communication between the server GUI and client GUI.

SYS-004

Description: The system shall provide the user option of using TCP/IP *or* BLE for communication.

SYS-005

Description: The system shall provide the capability to send and receive supported signal values from server to client.

2. Signal Requirements

SYS-010

Description: The system shall support a speed signal with a value range of 0–240 km/h.

SYS-011

Description: The system shall support a temperature signal with a value range of -60 – 60 °C.

SYS-012

Description: The system shall support a battery level signal with a value range of 0–100%.

SYS-013

Description: The system shall support a left blinker signal with the states active and inactive.

SYS-014

Description: The system shall support a right blinker signal with the states active and inactive.

SYS-015

Description: The system shall support a warning blinker signal with the states active and inactive.

SYS-016

Description: The system shall maintain a consistent representation of supported signal values between system components.

3. Server GUI Requirements

SYS-020

Description: The server GUI shall provide controls for entering and modifying supported numerical signal values.

SYS-021

Description: The server GUI shall provide controls for changing the state of the left, right, and warning blinkers.

SYS-022

Description: The server GUI shall not allow the left and right blinker signals to be active simultaneously.

SYS-023

Description: The server GUI shall indicate the current values and states of the supported signals.

SYS-024

Description: The server GUI shall prevent input of signal values outside their defined valid ranges.

4. Client GUI Requirements

SYS-030

Description: The client GUI shall display the current speed value.

SYS-031

Description: The client GUI shall display the current temperature value.

SYS-032

Description: The client GUI shall display the current battery level.

SYS-033

Description: The client GUI shall display the current left blinker status.

SYS-034

Description: The client GUI shall display the current right blinker status.

SYS-035

Description: The client GUI shall display the current warning blinker status.

SYS-036

Description: The client GUI shall provide an indication of the communication status.

5. Communication Requirements

SYS-040

Description: The system shall use TCP/IP for communication between the server GUI and client GUI.

SYS-041

Description: The system shall use UART for communication between the GUI applications and the server/client MCU:s.

SYS-042

Description: The system shall use Bluetooth Low Energy (BLE) for wireless communication between the server/client MCU:s.

SYS-043

Description: The system shall use a user defined communication protocol for transmitting signal values between system components.

SYS-044

Description: The communication interfaces shall support transmission and reception of the supported signal values.

SYS-045

Description: The system shall detect loss of communication between connected system components.

SYS-046

Description: The system shall provide a defined communication status indicating whether communication is available.

6. Communication Abstraction

SYS-050

Description: The system shall provide an abstract communication interface that separates application functionality from the underlying communication protocol.

SYS-051

Description: The communication interface shall provide a common interface for transmitting and receiving system data.

SYS-052

Description: The system shall provide a TCP/IP communication implementation through the communication interface.

SYS-053

Description: The system shall provide a UART communication implementation through the communication interface.

7. Reliability and Error Handling

SYS-060

Description: The system shall detect invalid or incomplete communication data.

SYS-061

Description: The system shall prevent invalid or incomplete communication data from being interpreted as valid signal values.

SYS-062

Description: The system shall provide a defined state for signals when their source is unavailable.

SYS-063

Description: The system shall support restoration of communication after a temporary communication failure.

8. Performance

SYS-070

Description: The system shall transmit updated signal values within the specified system update interval.

SYS-071

Description: The system shall display received signal values within the specified display update interval.

SYS-072

Description: The system shall support continuous communication between connected system components during normal operation.